Online Supplementary Material

A Synergy of Quantum-Inspired Pathfinding Algorithms, Reinforcement Learning, and Graph Neural Networks

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This document holds the diagnostic, superseded and index tables that the thesis cites as Supplement S.1–S.5 and Tables S1–S11. Section, table, figure and equation numbers without an S refer to the thesis. Artefact paths are relative to `https://github.com/Amir-Shahriari/Thesis_Project`.

S.1 Training-Budget Tiers and the Headline Scale

Metric (warm V1)	1× **(10 it)**	4× **(40 it)**
Strict goal-reach rate	22 / 25 (88 %)	**25 / 25 (100 %)**
Strict-win rate (warm < cold)	21 / 25 (84 %)	**24 / 25 (96 %)**
Mean cost ratio vs Dijkstra	17.3×	14.8×
Median cost ratio vs Dijkstra	10.5×	4.3×
Cells at ≤ 5× Dijkstra	—	13 / 25
Cold strict goal-reach (reference)	3 / 25 (12 %)	2 / 25 (8 %)

Table S1: 50 × 50 V1 warm results, 1× versus 4× tier, 5 seeds × 5 scenarios, batch 64. Training longer advances the perturbation schedule, so every across-tier comparison in this section is made on different realisations of each graph. Unmatched: all 22 comparable cells have a different Dijkstra reference, and the three cells unsolvable at 1× (`runs/eval_reachability_audit.json`) are connected at 4×. Reach supports no compute claim (three discordant against zero, McNemar $p = 0.25$); cost ratios are indicative only. Sources: `runs/sweep_50x50/aggregate_50x50.json`, `runs/sweep_50x50/sweep_v1_50x50_4x.json`.

Metric	**V1 (warm, 100×100)**	**V1 (warm, 25×25, from §4.3.1)**
Strict goal-reach rate	**6 / 25 (24 %)**	24 / 25 (96 %)
Reasonable-reach $\mathcal{R}_3$	0 / 25 (0 %)	36 %
Mean composite cost (reached)	2,248 ± 1,187	242 ± 197
Mean cost ratio vs Dijkstra	40.11	4.42
Mean per-trajectory wall-clock (ms)	99.4 ± 33.2	7.5 ± 10.5
Strict-win rate (warm < cold)	24 %	84 %
Paired t-test p-value	untestable[§]	0.0074

Table S2: V1 at 100 × 100 against the 25 × 25 baseline. Three cells are unsolvable at their evaluation states; over solvable cells warm reach is 6/22 (27.3 %). [§]Untestable: warm reaches 6 cells and cold 1, disjoint, so there are no pairs. The reach contrast is 6 discordant cells against 1, exact McNemar $p = 0.125$ (§4.3.3). Source: `runs/sweep_v1_on_100x100.json`.

Seed	Scenario id	1× reach	4× reach	4× cost	Cost ratio	Train (min)
42	seed191664964_s0	No	Yes	1,035	15.73	137.4
42	seed191664964_s1	No	No	—	—	35.1
42	seed191664964_s2	No	Yes	1,221	23.29	59.6
42	seed191664964_s3	No	No	—	—	58.9
1337	seed1173222464_s0	No	Yes	3,556	60.06	35.0
1337	seed1173222464_s1	No	Yes	3,530	59.83	58.9
1337	seed1173222464_s2	No	No	—	—	58.9
2024	seed518677876_s1	No	Yes	4,638	82.88	35.1
2024	seed518677876_s2	No	No	—	—	81.5
2024	seed518677876_s4	No	Yes	776	15.22	181.4
7	seed2029167941_s2	No	Yes	3,200	59.25	36.4
7	seed2029167941_s3	No	Yes	579	9.08	124.2
7	seed2029167941_s4	No	No	—	—	144.3
314159	seed1830621171_s0	No	Yes	4,662	84.26	35.0
314159	seed1830621171_s2	No	Yes	1,933	28.76	145.6
Aggregate		0 / 15 (0 %)	**10 / 15 (66.7 %)**	—	mean 43.84×	mean 81.8
95 % binomial CI on 4× reach rate		[0.384, 0.882]				
McNemar's test (1× vs 4×)		$p = 0.0020$				

Table S3: V1 long-train subset: 15 cells spanning all five seeds (two to four per seed), drawn from cells where the 1× budget failed; evaluated at iterations 10 and 40. The 4× instances are easier: 11 of 12 cells with a reference at both have a shorter optimum (sign test $p = 0.0063$; median $69.3 \rightarrow 60.0$), and one cell unsolvable at 1× is reached at 4×. The McNemar test is biased towards 4× by an unknown amount; the compute claim rests on the paired cold-start control. Sources: `runs/v1_long_train_subset_results.json`, `runs/v1_long_train_aggregate_15cell.json`.

S.2 Encoder-Geometry Diagnostic

Measurement	25×25 (d=64)	50×50 (d=128)
(A) Embedding geometry		
IsoScore (1 = isotropic)	0.044	0.034
Effective rank	20.1/64	25.6/128
Mean pairwise cosine	0.756	0.626
after mean-centring	0.021	0.019
Locality r_s, encoder†	0.124	0.057
Locality r_s, raw input features†	0.041	0.007
Locality r_s, *ground-truth coordinates*†	0.091	0.075
same probe, pure grid (E_{extra}=0)	0.978	0.980
(B) Retrieval gate, by embedding treatment		
Saturation ($\text{sim}^{\max} > 0.95$), raw	78.5 %	56.5 %
centred	12.3 %	9.2 %
whitened	0.05 %	0.43 %
Retrieval quality, raw	0.103	0.023
centred	0.102	0.025
whitened	0.090	0.022
Gate AUC, raw	0.584	0.636
(C) Metric probe: locality r_s by readout†		
Cosine (as deployed)	0.130	0.054
Learned ridge metric	0.224	0.154
Best learned metric (single best checkpoint)	0.287	0.195

Table S4: Encoder-geometry diagnostics for §4.6.1, 10 checkpoints per scale (five cells × warm/cold). **(A)** Geometry, including locality (Spearman r_s, embedding against hop distance). **(B)** Retrieval gate under four treatments (raw, mean-centred, whitened, percentile-calibrated): saturation, gate AUC, and retrieval quality (1.0 retrieves the nearest stored state, 0.0 a random one). **(C)** Metric probe: a learned metric (ridge, MLP) on frozen embeddings predicting hop distance. Warm against cold at 25×25: IsoScore 0.059 against 0.029, effective rank 23.7 against 16.5, mean cosine 0.666 against 0.845, warm better in all five scenarios; five pairs cannot reach significance (minimum Wilcoxon $p = 0.0625$), and none replicates at 50×50 (2/5, 3/5, 3/5 scenarios; $p \geq 0.31$). No claim is made that warm-starting mitigates encoder collapse. At 25×25 whitening cuts saturation from 78.5 % to 0.05 % (Wilcoxon $p = 0.002$) with no detectable change in retrieval quality ($p = 0.28$; centring $p = 0.92$). †Uninterpretable because shortcut edges compress hop distance (§3.2), so no conclusion is drawn. Sources: `runs/geometry_study_aggregate.json`, `runs/geometry_audit.json`, `runs/retrieval_gate.json`, `runs/metric_probe.json`; positive control via `scripts/analysis/locality_control.py` with $E_{\text{extra}} = 0$, values in `runs/thesis_support/item_c_locality_control.json`.

S.3 Cross-Distribution Transfer

Scenario	Grid	p_{deact}	V1 reach	V2 reach	V3 reach	V3 mean λ^{eff}
in-dist	100×100	0.30	2 / 10	3 / 10	2 / 10	0.321
new-seed	100×100	0.30	1 / 10	2 / 10	1 / 10	0.322
low-pert	100×100	0.15	2 / 10	2 / 10	3 / 10	0.312
high-pert	100×100	0.45	1 / 10	2 / 10	2 / 10	0.350
small-grid	50×50	0.30	1 / 10	1 / 10	2 / 10	0.337

Table S5: Cross-distribution transfer of V3 with the V1-trained agent and the V2 library, evaluated on five scenarios, ten queries per scenario (§4.5.2). V2 and V3, the expert library and the similarity gate are the superseded inference-time rules of Supplement S.4; inference-rule conditions are for cross-comparison only. Source: `runs/v3_transfer_benchmark.json`.

S.4 Superseded Inference-Augmentation Strand

Configuration	Agent	Library	λ rule	Goal-reach
V1 (baseline, no library)	V1-trained	—	—	11 / 25 (44 %)
V2-control (isolated mechanism effect)	V1-trained	yes	fixed 0.5	10 / 25 (40 %)
V2 published (agent + library, independent)	V2-trained	yes	fixed 0.5	8 / 25 (32 %)

Table S6: V2 on the 100×100 sweep. Superseded and retained for cross-comparison only; no claim in this thesis rests on the tables of this section. Two training-free inference-time mechanisms were layered over the trained V1 agent at 100×100. V2 blends a persistent expert library's recommendation at a fixed $\lambda_{\text{retr}} = 0.5$; V3 replaces the fixed coefficient with a similarity gate, $\lambda(s) = \lambda_{\max} \cdot \text{sim}^{\max}(s)$, where $\text{sim}^{\max}(s)$ is the highest cosine similarity between the current node's encoder embedding and any state embedding stored in the library, clipped below at zero, and λ^{eff} is the value of $\lambda(s)$ realised at a step, reported as a mean. V2-control uses the V1 checkpoint with the library at $\lambda_{\text{retr}} = 0.5$; V2 published uses a separately trained V2 agent. V1 and V2-control are at the matched 1,000-step budget (static regime of Table S10); V2 published comes from the 300-step headline sweep and is not budget-matched with them. Sources: `runs/unified_benchmark_100x100.json` (V1, static regime), `runs/sweep_v3_adaptive.json` (V2-control), `runs/sweep_v2_headline.json` (V2 published).

Configuration	Goal-reach	Mean cost (reached)	Mean cost ratio	Mean λ^{eff}
V1 (no library)	11 / 25 (44 %)	$8{,}934 \pm 4{,}858$	168.67	—
V2-control (fixed $\lambda = 0.5$)	10 / 25 (40 %)	$7{,}388 \pm 5{,}118$	132.16	0.50
V3 (adaptive λ, $\lambda_{\max} = 1.0$)	12 / 25 (48 %)	$8{,}097 \pm 4{,}092$	127.78	0.365

Table S7: V3 against V2-control and V1; one V1 checkpoint and the V2 library, inference rule varied, all at the matched 1,000-step budget. V2 reaches 10/25 and V3 12/25, results the static regime of Table S10 reproduces cell for cell; the 6/25 of the 300-step V1 sweep, on different evaluation states, is not a matched baseline. At the matched budget neither mechanism separates from V1 (b=1, c=2, $p = 1.0$ for V2; b=3, c=2, $p = 1.0$ for V3), nor does V3 from V2 (b=3, c=1, $p = 0.63$). Route quality is poor in all three: 132.2× Dijkstra for V2 and 127.8× for V3 against V1's 168.7×. Sources: `runs/sweep_v3_adaptive.json` (V2-control, V3), `runs/unified_benchmark_100x100.json` (V1, static regime).

$\lambda_{\max}$	Goal-reach	Mean cost (reached)	Mean λ^{eff}	Mean $\text{sim}^{\max}$
0.5	4 / 10	5,177	0.193	0.386
1.0	**6 / 10**	6,823	0.383	0.383
1.5	4 / 10	7,619	0.574	0.383
2.0	3 / 10	6,987	0.766	0.383

Table S8: V3 $\lambda_{\max}$ sensitivity on 10 cells (first two scenarios of each seed; not the first ten records of the sweep file). The $\lambda_{\max} = 1.0$ row is the headline run restricted to these cells, from `runs/sweep_v3_adaptive.json`; other rows from `runs/v3_lambda_ablation.json`.

Mode	Reach	Mean cost (reached)	Mean cost ratio	Mean λ^{eff}
V1 (no library)	6/10	5,344	84.0	—
V2 (fixed $\lambda_{\text{retr}} = 0.5$)	7/10	6,649	116.6	0.500
V3 (adaptive λ)	4/10	5,595	94.6	**0.974**

Table S9: V3 on the 10-cell long-train subset, one long-trained V1 agent per cell, library built at the evaluation state (matched encoder). McNemar V1 versus V3: $b = 2, c = 4, p = 0.6875$. The gate does not detect novelty: here it saturates at $\lambda^{\text{eff}} = 0.974$ and V3 reaches 4/10 cells against V1's 6/10, so the unsaturated gate of the headline sweep (mean $\lambda^{\text{eff}} = 0.365$) reflects encoder-state drift between library construction and evaluation; the representation itself is measured in §4.6.1. The evaluation uses a different perturbation realisation from Table S3, where the same agents reach all ten cells. Source: `runs/v3_long_train_subset_results.json`.

Regime (Δ)	Build	Goal-reach	Cost ratio	Latency (ms)	Survival
static	V1	11 / 25 (44 %)	168.7	541	—
	V2	10 / 25 (40 %)	132.2	5,962	—
	V3	**12 / 25 (48 %)**	127.8	5,011	—
	A*	24 / 25 (96 %)	1.36	21.4	—
	Cold	2 / 25 (8 %)	271.0	614	—
$\Delta = 50$	V1	11 / 25	165.3	612	1.00
	V2	8 / 25	125.1	5,955	1.00
	V3	11 / 25	130.1	6,227	1.00
	A**	24 / 25	1.36	27.3	n/a
	Cold	3 / 25	191.1	753	1.00
$\Delta = 30$	V1	11 / 25	208.8	629	1.00
	V2	10 / 25	123.5	6,047	1.00
	V3	10 / 25	138.9	6,696	1.00
	A**	24 / 25	1.36	20.9	n/a
	Cold	2 / 25	233.3	904	1.00
$\Delta = 15$	V1	13 / 25	161.7	763	1.00
	V2	9 / 25	101.2	7,474	1.00
	V3	11 / 25	190.2	7,682	1.00
	A**	24 / 25	1.36	15.8	n/a
	Cold	3 / 25	254.1	906	1.00

Table S10: Unified benchmark, $25 \times 5 \times 4 = 500$ evaluations (cells, conditions, regimes). Latency: mean per-trajectory wall-clock; survival: fraction of perturbation events after which the trajectory continued, 1 by construction (§3.7). *A* encounters zero perturbations in every regime (V1: 368–1,178) and its results are bit-identical across regimes, so its rows are only a reference for undisturbed path quality. Sources: `runs/unified_benchmark_100x100_aggregate.json`, `runs/unified_benchmark_100x100.json`.

S.5 Source Artefacts

Finding	Source artefact
25×25 sweep	`runs/sweep_phase3_final.json`
50×50 V1 sweep (1× tier, published June)	`runs/sweep_50x50/sweep_v1_50x50_1x.json`
Action-space A/B, reactive arm (50 × 50)	`runs/sweep_50x50_reactive_ctrl/`
Action-space A/B, masked arm (50 × 50)	`runs/sweep_50x50_masked_gpu/`
Action-space control, masked (100 × 100)	`runs/sweep_100x100_masked.json`
Masked multi-query control (50 × 50, 3 seeds)	`runs/fleet_fixed/reeval_seed*_50x50_mq_masked.json`

Finding	Source artefact
Per-run execution provenance (50×50 A/B arms only)	`provenance.json` (in each arm's run directory)
50×50 V1 sweep (4× tier)	`runs/sweep_50x50/sweep_v1_50x50_4x.json`
50×50 aggregate (gates, CI; McNemar stored asymptotic, exact values recomputed from per-cell files)	`runs/sweep_50x50/aggregate_50x50.json`
50×50 run record (config, S4-fix status)	`runs/sweep_50x50/REPORT.md`
Paired-test methods behind the gate scorecard	`runs/stats_reanalysis.json`
100×100 V1 sweep	`runs/sweep_v1_on_100x100.json`
100×100 V2 sweep + V2-control re-evaluation	`runs/sweep_v2_headline.json` (V2), `runs/sweep_v3_adaptive.json` (V2-control)
100×100 V3 sweep	`runs/sweep_v3_adaptive.json`
$\lambda_{\max}$ ablation	`runs/v3_lambda_ablation.json`
Cross-distribution transfer benchmark	`runs/v3_transfer_benchmark.json`
V1 long-train single cell	`runs/v1_long_train_results.json`
V1 long-train 15-cell subset (per-cell records)	`runs/v1_long_train_subset_results.json`
V1 long-train 15-cell aggregate (statistics, CI, McNemar)	`runs/v1_long_train_aggregate_15cell.json`
V3 long-train single-cell (legacy)	`runs/v3_long_train_results.json`
V3 long-train 10-cell subset (per-cell records)	`runs/v3_long_train_subset_results.json`
V3 long-train 10-cell aggregate (statistics)	`runs/v3_long_train_subset_aggregate.json`
Within-trajectory perturbation (25×25, conditions A/B/C)	`runs/v3_realtime_25x25_aggregate.json`
Unified comparison	`runs/unified_benchmark_100x100.json`, `runs/unified_benchmark_100x100_aggregate.json`
Unified sanity check	`runs/unified_benchmark_sanity.json`
Pareto plot data	`runs/unified_pareto_100x100.csv`
V1 perturbed-copies check (P1 gate)	`runs/perturbation_phase4.json`
Cold-start 4× control (per-cell + aggregate)	`runs/cold_4x_control_results.json`, `runs/cold_4x_control_aggregate.json`
Reward-design ablation (30 runs)	`runs/reward_ablation/reward_ablation_results.json`
Reward-shaping control (five arms, 25×25)	`runs/shaping_control/aggregate.json`, `runs/shaping_control/lambda_*.json`
Eval-state reachability audit	`runs/eval_reachability_audit.json`
Consolidated provenance inventory	`runs/INVENTORY.md`
Demonstration-source ablation (25×25), run log	`runs/demo_source_ablation.log`

Finding	Source artefact
Demonstration-source ablation, partial join (9/50)	`runs/demo_source_ablation_partial.json`
Per-arm demonstration diversity (25×25)	`runs/traces_25x25/*/*/seeding_diversity.json`
Sample efficiency (episodes-to-threshold, 25×25)	`runs/learning_curves_25x25.json`
Multi-query training, per seed (25×25)	`runs/fleet_fixed/fleet_results_seed*_25x25.json`
Multi-query held-out re-evaluation (150 queries)	`runs/fleet_fixed/reeval_seed*_25x25.json`
Goal-relative head, per seed	`runs/fleet_fixed/reeval_seed*_25x25_gr.json`
Fixed-code replication log (commit 4a231ba)	`runs/fleet_fixed/STATUS.txt`
Hindsight experience replay (5 seeds)	`runs/fleet_her_fixed/reeval_seed*_25x25_her.json`
Multi-query training, per seed (50 × 50)	`runs/fleet_fixed/fleet_results_seed*_50x50_mq.json`
Action-space A/B, reactive arm (25 × 25)	`runs/sweep_phase3_unmasked_ctrl.json`
Action-space A/B, masked arm (25 × 25)	`runs/sweep_phase3_masked.json`
Action-space A/B, both arms re-run on one recorded stack (25 × 25)	`runs/thesis_support/item_g_ab25_rerun/`
Termination-reason probe (reactive)	`runs/traces_reactive_probe/`
Masked replication, per-episode traces	`runs/traces_25x25_masked/`
Masked multi-query control (25 × 25, 5 seeds)	`runs/fleet_fixed/reeval_seed*_25x25_masked.json`
Encoder geometry, per checkpoint	`runs/geometry_audit.json`
Encoder geometry, aggregate (A/B/C studies)	`runs/geometry_study_aggregate.json`
Retrieval-gate treatments	`runs/retrieval_gate.json`
Learned-metric probe	`runs/metric_probe.json`
Pre-registered persistence ablation (50 × 50, masked)	`runs/persistence_ablation_50x50/aggregate.json`, `runs/persistence_ablation_50x50/cells_seed_*.json`
Persistence ablation pre-registration	`runs/persistence_ablation_50x50/PREREGISTRATION.md`
Demonstration-source ablation, traced full-pool and cold reference (25 × 25)	`runs/sweep_phase3_traced.json`
Multi-query held-out re-evaluation (50 × 50)	`runs/fleet_fixed/reeval_seed*_50x50_mq.json`
Exploration statistics on the 25 × 25 preset graph (regenerated)	`runs/thesis_support/item_a_preset_exploration_rules.json`

Finding	Source artefact
Within-trajectory Dijkstra-denominator inflation (regenerated)	`runs/thesis_support/item_b_realtime_dijkstra_inflation.json`
Locality positive control and hop IQR (regenerated)	`runs/thesis_support/item_c_locality_control.json`
Reward calibration and optimal returns (regenerated)	`runs/thesis_support/item_d_reward_calibration.json`
Exact graph diameters (regenerated)	`runs/thesis_support/item_e_exact_diameters.json`
Within-trajectory re-encode cost (regenerated)	`runs/thesis_support/item_f_reencode_timing.json`
Trainable parameter counts (regenerated)	`runs/thesis_support/item_h_param_counts.json`
Random-policy reach floor (regenerated)	`runs/thesis_support/item_i_random_policy_floor.json`
Pre-registered goal-adjacent seeding ablation (50×50, masked)	`runs/goaladj_ablation_50x50/aggregate.json`, `runs/goaladj_ablation_50x50/cells_seed_*.json`, `runs/goaladj_ablation_50x50/PREREGISTRATION.md`

Table S11: Source artefacts, relative to the project repository. Rows marked (regenerated) were produced in October 2026: three from the recorded analysis scripts, and the Dijkstra-inflation, exact-diameter and re-encode-timing artefacts from new deterministic analyses, all kept in `runs/thesis_support/scripts/`. Artefacts from two working trees and successive code revisions are consolidated in `runs/`, indexed by `runs/INVENTORY.md`, which records origin and superseded or incomplete status. Only artefacts produced after the protocol of §3.8.1 carry a `provenance.json` (`torch` build, device, compute capability, revision); earlier stacks cannot be established retrospectively. Superseded and not cited: `runs/sweep_50x50_masked/` (stack unrecorded). Incomplete: the demonstration-source ablation's aggregation aborted on a reproducibility guard, and `runs/demo_source_ablation_partial.json` holds only 9 seed-42 cells from a separate re-run, whose quantum-only costs differ from the logged run on three of four cells; the figures of §4.5.1 are recovered from the run log by `scripts/analysis/ablation_from_log.py`. Other artefacts in the repository are not evidence: the superseded 50×50 masked run (Table B.4); and perturbation-level diagnostics, a multiplier sweep on demonstration checkpoints (five cells per scale) and three density-robust retraining attempts (one 50×50 cell), outside the evaluation grid and not reported in Chapter 4. The reward-shaping control (§4.4.2, Table B.8) is the one declared exception to the provenance rule.